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From Planning to Public Benefit: How Open Data Improves Your Research

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Introduction

Research is increasingly a team sport, with collaborative research having the highest impact (Wuchty et al., 2007). Simultaneously, a significant 'crisis' of reproducibility in research has become apparent (Baker, 2016). An increasing demand for transparency, reproducibility, and collaboration has propelled open data to the forefront of scientific and policy agendas (Baker, 2016; Murray-Rust, 2008). Making data open, such that it can be freely used, modified, and

shared by possibly anyone for any purpose has the potential to lower barriers to access, facilitate innovation, support evidence-based policy, and strengthen public trust in research (Open Knowledge Foundation ; Piwowar & Vision, 2013).

This paper explores key dimensions of open data, including principles and definitions, data management and planning, long-term retention and sustainability, and the pathways through which data is made publicly available. It also addresses challenges and future trends, offering guidance for researchers and institutions seeking to integrate open data practices responsibly.

Understanding Open Data

Open data is defined as data that potentially anyone can freely access, use, and share for any purpose. The definition of open data aligns closely with similar definitions in software development, where 'open-sourced' software contrasts with closed-source/proprietary software and instead enables anyone to inspect and re-use the code. The FAIR principles (Findable, Accessible, Interoperable, and Reusable) emphasise openness while also ensuring re-usability and quality (Wilkinson et al., 2016). While FAIR data are not necessarily open, open data initiatives often adopt FAIR-aligned practices to maximise reuse.

Applications and Benefits

Open data initiatives have impacted all domains of research and innovation, and the benefits of open data are clear. There is no doubt that the pace and impact of scientific innovation have been enhanced by open access; genetic sequencing data through initiatives such as the Human Genome Project and GenBank nicely highlight this (Benson et al., 2013; Collins & McKusick, 2001; Tenopir et al., 2011). While open data sharing is not sufficient to ensure reproducibility (Hardwicke et al., 2018; Obels et al., 2020), it is necessary for enabling verification and cumulative knowledge building (Peng, 2011). For example, open geospatial data are now routinely used in disaster response (Goodchild & Glennon, 2010). Through enabling innovation in industry and civil society, the economic value of data is transformed when data is made open (Manyika et al., 2013).

Societal change can also be driven by open data, supported by policy change. For example, Mexico, a country that

consistently ranks highly in open data metrics, leverages open data to hold schools accountable and reduce maternal mortality (Davies et al., 2019; Domínguez-Domínguez et al., 2023; Young & Verhulst, 2016). Relatedly, increased transparency and accountability can improve public trust in research and policymaking (National Academies of Sciences, Engineering, and Medicine, 2018; Wiencierz & Luenich, 2022).

Ethical and Legal Considerations

Open data are not universally applicable. Privacy concerns, intellectual property, and cultural factors require careful navigation. However, these concerns are not normally barriers for adopting open data sharing. For example, the CARE (Collective Benefit, Authority to Control, Responsibility, and Ethics) principles provide guidelines for Indigenous data governance, which can supplement and work alongside the FAIR principles to ensure data sovereignty when working with open data relating to Indigenous communities. Licenses such as Creative Commons or Open Data Commons clarify permissible uses and obligations, enabling responsible sharing while protecting rights holders (Creative Commons ; Open Knowledge Foundation).

Data Management and Planning

Effective open data begins with understanding the data lifecycle, which includes data collection, processing, documentation, storage, sharing, and reuse. Each stage requires deliberate planning to ensure data remains interpretable and accessible. As a result, successful implementation of open data practices starts before data collection.

Data Management Plans (DMPs) are increasingly mandated by funders and journals (Smale et al., 2020). Feasible open data sharing requires consideration of the types and formats of data to be collected, metadata standards, and documentation associated with the data, storage, backup, and security of data, along with plans for sharing, licensing, and long-term preservation. A robust DMP will address each of these (Michener, 2015).

Integrating openness into the entire project lifecycle from proposal drafting to project closure reduces downstream

barriers, while ensuring compliance with emerging open science mandates, while maximising the impact of the work (Piwowar & Vision, 2013). Put simply, it's easier to design a good DMP than to attempt to identify solutions when the need to share arises after the fact.

Digital tools such as DMPonline (UK, EU) and DMP Tool (US) streamline data planning, along with depositing and sharing data management plans (Sallans & Donnelly, 2012). As a result, good data management plans are themselves increasingly shared as open data enabling all those working with data to implement robust data management planning. Increasingly, universities and research consortia also provide training and templates to support researchers in aligning with best practices (Tenopir et al., 2011).

Documentation of Metadata

Metadata is structured descriptive information about data. Data re-use requires appropriate metadata as without adequate documentation, data may be technically 'open' but practically unusable. Metadata describes the data being shared, and provides the information required to interrogate and re-use the data. Standards, such as the Dublin Core for general metadata, or domain-specific standards such as the Darwin Core for biodiversity data, enable interoperability, cross-referencing, and computational discovery of open data (Dublim Core Metadata Initiative ; Wieczorek et al., 2012). For example, biodiversity metadata may contain the date, species scientific name, and the latitude-longitude coordinates in a standard format to ensure that others can interpret which species was found where and when. The FAIRsharing resource, developed by a team of international stakeholders across science, publishing, industry, and research funding, unites data reporting and metadata standards across disciplines (Sansone et al., 2019).

Long-term Retention and Sustainability

The value of research data often extends well beyond the life cycle of the original project. Longitudinal studies, replication efforts, and historical baselines (such as in climate research) depend on sustained access to archived data (Kitchin, 2014). Long-term data stewardship requires appropriate planning and budgeting. Beyond initial storage, ongoing metadata

maintenance, format updates (e.g., migrating files from obsolete formats into current, widely supported formats), repository management, and technical support for data access must be considered (Beagrie & Houghton, 2014).

Best practices emphasise redundancy (multiple copies across locations) and format migration to prevent obsolescence (Phillips et al., 2013). Budgeting for these expenses during grant planning mitigates future risks of data loss or 'orphaned' datasets, while also demonstrating to reviewers of funding applications that you have fully considered maximising the lifespan and impact of your data. Analysis of the cost of data storage demonstrates remarkable value, as re-generating lost or orphaned data will always be more expensive than storage (Piwowar et al., 2011).

Environmental Considerations

Data storage carries a carbon footprint, as a substantial proportion of the significant energy consumed by data centres is for data storage (Jones, 2018; Tomes & Altiparmak, 2017). While the energy impact of data management may be a minor part of an astrophysics study or a clinical trial (Griffiths et al., 2024; Martin et al., 2022), in other disciplines data storage can be a significant proportion of the environmental impact of the study (Mersy & Krishnan, 2024). Sustainable open data strategies may include selective archiving, data compression, and transitioning to low-carbon storage solutions.

Pathways to Public Availability

Long-term data storage is often achieved through institutional servers or cloud-based systems. Many domain-specific repositories exist to enable long-term retention of specific data types (e.g. proteomics: ProteomeXchange, Human genotype-phenotype: dbGAP), along with general-purpose repositories (e.g. Zenodo, Figshare, or the Open Science Framework) (Amorim et al., 2015). Depositing data in trusted repositories ensures persistence, discoverability, and proper citation (Piwowar & Vision, 2013). Repositories assign persistent identifiers such as DOIs (Digital Object Identifiers), and often adhere to community-specific metadata standards to ensure that data can be re-used.

While sensitive datasets require controlled-access agreements if the data cannot be sufficiently anonymised, sharing data as openly as possible saves time for both data-generating researchers and for researchers looking to benefit from open data (Lowndes et al., 2017). Making data freely available to download should be combined with appropriate licensing to clarify permissible reuse. For example, the CC BY licenses allow free reuse with attribution, while CC0 dedicates data to the public domain (Creative Commons).

Data journals (e.g. Scientific Data or Earth System Science Data) and papers (articles describing datasets and methodologies) provide scholarly recognition for data sharing movements and facilitate contextual understanding (Candela et al., 2015). Increasingly, open data publishing is encouraged by major publishing groups, along with a mandate for data availability statements. Many publishers require raw data sharing as a condition of publication, and some even as a condition of peer review (Miyakawa, 2020).

While the publication of original computational code is often encouraged by publishers, and may contribute positively to research metrics (Kang et al., 2023), the true power of open data lies in the simultaneous availability and linking of raw data and code to be able to reproduce the analysis (Cao et al., 2023; Peng, 2011). With the increasing availability of resources for storing both raw and processed data, both should be made open where possible. For example, in cancer research raw genetic data may be deposited in The National Center for Biotechnology Information Gene Expression Omnibus while the code for processing this raw data may be made available through Zenodo, with the processed data deposited in a repository such as the cBio Cancer Genomics Portal (Cerami et al., 2012). In this way, researchers looking to reuse, reproduce, or interrogate data can do so without unnecessary duplication of the original study.

Challenges and Future Directions

Global trends favour openness: the European Union's Horizon Europe programme and the United States federal data strategy exemplify a shift toward 'open by default'. Future frameworks must balance open science ideals with ethical and security concerns.

Heterogeneity in formats, metadata standards, and repository capabilities impedes interoperability. Cross-disciplinary collaboration and consensus building must combine with investment in robust infrastructure to overcome these barriers.

Despite policy shifts, cultural resistance to open data persists. Concerns include loss of competitive advantage, fear of misinterpretation, and inadequate recognition for data curation (Tenopir et al., 2011). Incentive structures that reward data sharing through citations or funding metrics are essential. Despite growing recognition of the value of open science, access to clinical and health data remains fragmented and heavily restricted. Cancer research in particular highlights the challenges: data are often siloed across regional and national systems, subject to complex governance frameworks, and can take months to access, creating barriers to reuse and slowing progress toward patient benefit (Asiimwe et al., 2021; Denton et al., 2021).

As artificial intelligence (AI) and machine learning increasingly utilise open datasets, the value of data will expand exponentially. Breakthroughs such as Alphafold (an AI system developed by DeepMind that predicts protein 3D structures from amino acid sequences) were made possible by the field of structural biology embracing data sharing via open access licenses and standardised data formats (Jumper et al., 2021; wwPDB consortium, 2019). However, this also amplifies risks of bias and misuse, for example when health datasets underrepresent certain populations, leading to AI tools that perform poorly or unfairly in those groups. This underscores the need for governance and frameworks that ensure responsible openness (Leslie, 2019). Federated data sharing allows researchers to analyse sensitive datasets without centralizing them, balancing openness with privacy. Such approaches can, for example, enable AI models to be trained while sensitive clinical data remains stored where it was generated, in an approach termed federated learning (van Genderen et al., 2024).

Conclusion

Open data represents a paradigm shift in research and governance, offering unprecedented opportunities for transparency, reproducibility, and innovation. Realising these benefits requires careful planning, robust

documentation, and sustainable stewardship practices. By integrating ethical, financial, and environmental considerations, researchers and institutions can ensure that open data remains a public good for generations to come.

References

Amorim, R. C., Castro, J. A., Da Silva, J. R., & Ribeiro, C. (2015). A comparative study of platforms for research data management: interoperability, metadata capabilities and integration potential. In *New Contributions in Information Systems and Technologies: Volume 1* (pp. 101–111). Springer. https://doi.org/10.1007/978-3-319-16486-1_10

Asiimwe, R., Lam, S., Leung, S., Wang, S. Z., Wan, R., Tinker, A., McAlpine, J. N., Woo, M. M. M., Huntsman, D. G., & Talhouk, A. (2021). From biobank and data silos into a data commons: convergence to support translational medicine. *Journal of Translational Medicine*, *19*(1), 493. https://doi.org/10.1186/s12967-021-03147-z

Baker, M. (2016). 1,500 scientists lift the lid on reproducibility. In: Nature Publishing Group UK London.

Beagrie, N., & Houghton, J. W. (2014). The Value and Impact of Data Sharing and Curation A synthesis of three recent studies of UK research data centres. https://doi.org/10.1087/095315104322710197

Benson, D. A., Cavanaugh, M., Clark, K., Karsch-Mizrachi, I., Lipman, D. J., Ostell, J., & Sayers, E. W. (2013). GenBank. *Nucleic Acids Res*, *41*(Database issue), D36–42. https://doi.org/10.1093/nar/gks1195

Candela, L., Castelli, D., Manghi, P., & Tani, A. (2015). Data journals: A survey. *Journal of the Association for Information Science and Technology*, *66*(9), 1747–1762. https://doi.org/10.1002/asi.23358

Cao, H., Dodge, J., Lo, K., McFarland, D. A., & Wang, L. L. (2023). The rise of open science: Tracking the evolution and perceived value of data and methods link-

sharing practices. *arXiv preprint arXiv:2310.03193*. https://doi.org/10.1109/icce.2010.5418689

Cerami, E., Gao, J., Dogrusoz, U., Gross, B. E., Sumer, S. O., Aksoy, B. A., Jacobsen, A., Byrne, C. J., Heuer, M. L., Larsson, E., Antipin, Y., Reva, B., Goldberg, A. P., Sander, C., & Schultz, N. (2012). The cBio cancer genomics portal: an open platform for exploring multidimensional cancer genomics data. *Cancer Discov*, *2*(5), 401–404. https://doi.org/10.1158/2159-8290.CD-12-0095

Collins, F. S., & McKusick, V. A. (2001). Implications of the Human Genome Project for medical science. *Jama*, *285*(5), 540–544. https://doi.org/10.1001/jama.285.5.540

Creative Commons Retrieved August 4 from https://creativecommons.org/licenses/

Davies, T., Walker, S. B., Rubinstein, M., & Perini, F. (2019). *The state of open data: Histories and horizons*. African Minds. https://doi.org/10.1007/978-3-030-44556-0_102

Denton, N., Molloy, M., Charleston, S., Lipset, C., Hirsch, J., Mulberg, A. E., Howard, P., & Marsh, E. D. (2021). Data silos are undermining drug development and failing rare disease patients. *Orphanet J Rare Dis*, *16*(1), 161. https://doi.org/10.1186/s13023-021-01806-4

Domínguez-Domínguez, R., Alférez, G. H., González-Mejia, V., & Donías, N. (2023). Data Science Application for Creation of Maternal Morbidity and Mortality Predictive Software. In: Sep.

Dublim Core Metadata Initiative Retrieved August 4 from https://www.dublincore.org/specifications/dublin-core/dcmi-terms/

Goodchild, M. F., & Glennon, J. A. (2010). Crowdsourcing geographic information for disaster response: a research frontier. *International Journal of Digital Earth*, *3*(3), 231–241. https://doi.org/10.1080/17538941003759255

Griffiths, J., Fox, L., Williamson, P. R., & Low Carbon Clinical Trials, G. (2024). Quantifying the carbon footprint of clinical trials: guidance development and case studies. *BMJ open*, *14*(1), e075755. https://doi.org/10.1136/bmjopen-2023-075755

Hardwicke, T. E., Mathur, M. B., MacDonald, K., Nilsonne, G., Banks, G. C., Kidwell, M. C., Mohr, A. H., Clayton, E., Yoon, E. J., Tessler, M. H., Lenne, R. L., Altman, S., Long, B., & Frank, M. C. (2018). Data availability, reusability, and analytic reproducibility: evaluating the impact of a mandatory open data policy at

the journal. *Royal Society open science*, *5*(8), 180448. https://doi.org/10.1098/rsos.180448

Jones, N. (2018). How to stop data centres from gobbling up the world's electricity. *Nature*, *561*(7722), 163–166. https://doi.org/10.1038/d41586-018-06610-y

Jumper, J., Evans, R., Pritzel, A., Green, T., Figurnov, M., Ronneberger, O., Tunyasuvunakool, K., Bates, R., Zidek, A., Potapenko, A., Bridgland, A., Meyer, C., Kohl, S. A. A., Ballard, A. J., Cowie, A., Romera-Paredes, B., Nikolov, S., Jain, R., Adler, J.,...Hassabis, D. (2021). Highly accurate protein structure prediction with AlphaFold. *Nature*, *596*(7873), 583–589. https://doi.org/10.1038/s41586-021-03819-2

Kang, D., Kang, T. Y., & Jang, J. (2023). Papers with code or without code? Impact of GitHub repository usability on the diffusion of machine learning research. *Information Processing & Management*, *60*(6), 103477. https://doi.org/10.1016/j.ipm.2023.103477

Kitchin, R. (2014). *The data revolution: Big data, open data, data infrastructures and their consequences*. Sage. https://doi.org/10.4135/9781473909472

Leslie, D. (2019). Understanding artificial intelligence ethics and safety. *arXiv preprint arXiv:1906.05684*. https://doi.org/10.2139/ssrn.3403301

Lowndes, J. S. S., Best, B. D., Scarborough, C., Afflerbach, J. C., Frazier, M. R., O'Hara, C. C., Jiang, N., & Halpern, B. S. (2017). Our path to better science in less time using open data science tools. *Nature ecology & evolution*, *1*(6), 0160. https://doi.org/10.1038/s41559-017-0160

Manyika, J., Chui, M., Groves, P., Farrell, D., Van Kuiken, S., & Almasi Doshi, E. (2013). *Open Data: Unlocking Innovation and Performance with Liquid Information*. https://www.mckinsey.com/~/media/McKinsey/Business%20Functions/McKinsey%20Digital/Our%20Insights/Open%20data%20Unlocking%20innovation%20and%20performance%20with%20liquid%20information/MGI_Open_data_FullReport_Oct2013.ashx

Martin, P., Brau-Nogué, S., Coriat, M., Garnier, P., Hughes, A., Knödlseder, J., & Tibaldo, L. (2022). A comprehensive assessment of the carbon footprint of an astronomical institute. *Nature Astronomy*, *6*(11), 1219–1222. https://doi.org/10.1038/s41550-022-01771-3

Mersy, G., & Krishnan, S. (2024). Toward a life cycle assessment for the carbon footprint of data. *ACM SIGENERGY Energy Informatics Review*, *4*(5), 25–33. https://doi.org/10.1145/3727200.3727206

Michener, W. K. (2015). Ten Simple Rules for Creating a Good Data Management Plan. *PLoS Comput Biol*, *11*(10), e1004525. https://doi.org/10.1371/journal.pcbi.1004525

Miyakawa, T. (2020). No raw data, no science: another possible source of the reproducibility crisis. *Molecular brain*, *13*(1), 24. https://doi.org/10.1186/s13041-020-0552-2

Murray-Rust, P. (2008). Open Data in science. *Serials Review*, *34*(1), 52–64. https://doi.org/10.1016/j.serrev.2008.01.001

National Academies of Sciences, Engineering, and Medicine. (2018). *Open science by design: Realizing a vision for 21st century research*. https://doi.org/10.17226/25116

Obels, P., Lakens, D., Coles, N. A., Gottfried, J., & Green, S. A. (2020). Analysis of Open Data and Computational Reproducibility in Registered Reports in Psychology. *Advances in Methods and Practices in Psychological Science*, *3*(2), 229–237. https://doi.org/10.1177/2515245920918872

Open Knowledge Foundation Retrieved August 4 from https://opendefinition.org/od/2.1/en/

Peng, R. D. (2011). Reproducible research in computational science. *Science*, *334*(6060), 1226–1227. https://doi.org/10.1126/science.1213847

Phillips, M., Bailey, J., Goethals, A., & Owens, T. (2013). The NDSA levels of digital preservation: Explanation and uses. Archiving conference,

Piwowar, H. A., & Vision, T. J. (2013). Data reuse and the open data citation advantage. *PeerJ*, *1*, e175. https://doi.org/10.7717/peerj.175

Piwowar, H. A., Vision, T. J., & Whitlock, M. C. (2011). Data archiving is a good investment. *Nature*, *473*(7347), 285–285. https://doi.org/DOI 10.1038/473285a

Sallans, A., & Donnelly, M. (2012). DMP online and DMPTool: Different strategies towards a shared goal. *International Journal of Digital Curation*, *7*(2), 123–129. https://doi.org/10.2218/ijdc.v7i2.235

Sansone, S. A., McQuilton, P., Rocca-Serra, P., Gonzalez-Beltran, A., Izzo, M., Lister, A. L., Thurston, M., & Community, F. (2019). FAIRsharing as a community

approach to standards, repositories and policies. *Nature biotechnology*, *37*(4), 358–367. https://doi.org/10.1038/s41587-019-0080-8

Smale, N. A., Unsworth, K., Denyer, G., Magatova, E., & Barr, D. (2020). A review of the history, advocacy and efficacy of data management plans. *International Journal of Digital Curation*, *15*(1), 30–30. https://doi.org/10.2218/ijdc.v15i1.525

Tenopir, C., Allard, S., Douglass, K., Aydinoglu, A. U., Wu, L., Read, E., Manoff, M., & Frame, M. (2011). Data sharing by scientists: practices and perceptions. *PloS one*, *6*(6), e21101. https://doi.org/10.1371/journal.pone.0021101

Tomes, E., & Altiparmak, N. (2017). A comparative study of HDD and SSD RAIDs' impact on server energy consumption. 2017 IEEE International Conference on Cluster Computing (CLUSTER),

van Genderen, M. E., Cecconi, M., & Jung, C. (2024). Federated data access and federated learning: improved data sharing, AI model development, and learning in intensive care. *Intensive Care Medicine*, *50*(6), 974–977. https://doi.org/10.1007/s00134-024-07408-5

Wieczorek, J., Bloom, D., Guralnick, R., Blum, S., Doring, M., Giovanni, R., Robertson, T., & Vieglais, D. (2012). Darwin Core: an evolving community-developed biodiversity data standard. *PloS one*, *7*(1), e29715. https://doi.org/10.1371/journal.pone.0029715

Wiencierz, C., & Luenich, M. (2022). Trust in open data applications through transparency. *New media & society*, *24*(8), 1751–1770. https://doi.org/10.1177/1461444820979708

Wilkinson, M. D., Dumontier, M., Aalbersberg, I. J., Appleton, G., Axton, M., Baak, A., Blomberg, N., Boiten, J. W., da Silva Santos, L. B., Bourne, P. E., Bouwman, J., Brookes, A. J., Clark, T., Crosas, M., Dillo, I., Dumon, O., Edmunds, S., Evelo, C. T., Finkers, R.,...Mons, B. (2016). The FAIR Guiding Principles for scientific data management and stewardship. *Sci Data*, *3*(1), 160018. https://doi.org/10.1038/sdata.2016.18

Wuchty, S., Jones, B. F., & Uzzi, B. (2007). The increasing dominance of teams in production of knowledge. *Science*, *316*(5827), 1036–1039. https://doi.org/10.1126/science.1136099

wwPDB consortium. (2019). Protein Data Bank: the single global archive for 3D macromolecular structure data. *Nucleic acids research*, *47*(D1), D520–D528. https://doi.org/10.1107/s205327332209413x

Young, A., & Verhulst, S. (2016). Mexico's Mejora Tu Escuela: Empowering citizens to make data-driven decisions about education. *Open Data's Impact*. https://doi.org/10.1145/3725979